

T0 Theory: Document 185

Time as the Embedding Price of Mathematics

Factorisation, Quantum Computers, and the Physical Limit of RSA Decomposition

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April 2026

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Abstract

Factorisation is a timeless identity in pure mathematics: $N = p \cdot q$ exists without process, without runtime, without error. As soon as this identity is physically realised — whether on a quantum processor, a classical server, or the vacuum field itself — information must flow. Flow requires time. And time, defined in FFGFT by the fundamental revolution time $T_{\text{rev}} = \xi \cdot \ell_P / c$, accumulates phase errors.

This document shows: it is not insufficient qubit count or poor hardware that limits RSA decomposition by quantum computers, but the intrinsic time-phase coupling of physical spacetime. There is a scaling limit beyond which even the best future quantum computer fails — geometrically forced, not technically circumventable.

At the same time, the same geometry yields a positive result: error correction can be massively improved when one accounts for the fact that the vacuum field has preferred resonance points. These are not artefacts — they are the stable torus modes to which the physical system naturally returns.

.1 The Paradox of Timelessness

Two Worlds: Mathematics and Physics

In pure mathematics, factorisation $N = p \cdot q$ is a **timeless identity**. There is no process — only a static fact, simultaneously and always true. The prime p does not need to be searched; it is already contained in N without ever having been “placed” there.

In physical reality this is different. Whether on Google’s Willow processor, a classical server, or the human brain: every computation requires that information flows from one place to another. Flow means time. Time means phase drift.

The Central Distinction

Mathematics: $N = p \cdot q$ — timeless, error-free, without process.

Physics: To *extract* p and q from N , a measurement must take place — a period r must be found in the feedback $f(x) = a^x \bmod N$. This requires runtime $t = n \cdot T_{\text{rev}}$. Runtime accumulates phase errors. Beyond a critical N , the error exceeds the coherence threshold before the resonance is found.

Static Knowledge vs. Dynamic Process

An important distinction: known prime numbers do not need to be searched again. They exist as already identified, stable resonance points — entered in tables, stored in databases. This **knowledge** is static: retrieving a known prime costs no runtime.

But the *use* of these known primes to decompose a new, unknown number N is not a static comparison — it is a **dynamic process**. Using known primes shortens the search, but verifying the phase coincidence between a candidate and N requires physical runtime.

Knowledge vs. Process in FFGFT Language

In the language of torus resonances (Doc. 183):

Known prime p : Stored winding number — a topologically stable node that exists without further effort.

Factorising N : Checking whether the winding number of N can be written as the product of two known winding numbers $p \cdot q$. This check requires the quantum processor to physically establish the phase coincidence between p , q , and N — in real time, with accumulated phase error.

.2 Factorisation as Resonance Search in Time

Period Finding as a Physical Process

Shor’s algorithm (and T0-Shor, Doc. 075) solve factorisation through **period finding**: one searches for the period r of the function $f(x) = a^x \bmod N$. This period encodes the prime factors:

$$a^r \equiv 1 \pmod{N} \Rightarrow a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod{N} \quad (1)$$

The sought resonance frequency is $\nu_r = 1/r$. The required resolution is (Doc. 075):

$$\Delta f_{\text{required}} \approx \frac{f_0}{r^2} \quad (2)$$

For cryptographically relevant numbers with $r \approx N$:

Key length	Required precision ϵ	Status
27-bit	$\epsilon \approx 10^{-16}$	64-bit hardware sufficient
56-bit	$\epsilon \approx 10^{-34}$	128-bit hardware sufficient
RSA-1024	$\epsilon \approx 10^{-617}$	physically impossible
RSA-2048	$\epsilon \approx 10^{-1234}$	physically impossible

Table 1: Precision barrier for factorisation (after Doc. 075)

Why Runtime Accumulation Is Unavoidable

In FFGFT, time is not an external clock but the accumulated travel time of the field through its own geometry — measured in multiples of T_{rev} (Doc. 183):

$$T_{\text{rev}} = \frac{\xi \cdot \ell_P}{c} \approx 7.2 \times 10^{-48} \text{ s} \quad (3)$$

To find the period r , the system must complete r revolutions — a total runtime of $t = r \cdot T_{\text{rev}}$. At each revolution a phase error ϵ accumulates. After r cycles the total error is $r \cdot \epsilon$.

Coherence collapses as soon as the phase error exceeds $\pi/2$:

$$r_{\text{crit}} \approx \frac{\pi}{2\epsilon} \quad (4)$$

For RSA keys, $r \approx N$ is enormous. The required precision per step scales as $\epsilon \lesssim \pi/(2N)$ — which for RSA-1024 demands the physically impossible precision $\epsilon \approx 10^{-617}$.

Runtime Accumulation as a Law of Nature

This is not a technical constraint that better hardware can overcome. It is a consequence of the structure of physical reality:

$$\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi) \quad (5)$$

Every computational step is one revolution on the torus. Every revolution accumulates $\Delta\phi = 2\pi \cdot \xi$. For $r \approx N$ revolutions at RSA key lengths, the accumulated error $r \cdot \Delta\phi \approx N \cdot 2\pi \cdot \xi$ is far beyond the coherence threshold.

.3 Why the Quantum Computer Still Fails

The Threshold Theorem and Its Limits

Standard quantum information theory appeals to the **threshold theorem**: when the error rate per gate falls below a critical threshold, quantum error correction (QEC) can correct errors faster than they arise. The logical state then lives theoretically indefinitely.

FFGFT reveals a fundamental limit to this argument:

QEC Is Itself a Recursive Process

QEC requires runtime — every error correction is itself a sequence of gate operations that costs T_{rev} . The correction adds new phase errors to the system, which must themselves be corrected.

For small N (short periods, few cycles) the correction outweighs the error — the threshold theorem holds. For large N (RSA keys) the correction itself becomes part of

the problem: the recursion depth of error correction is so large that its phase error assumes the same character as the original noise.

The standard literature's argument — "error correction defeats time" — holds in the regime of short periods. In the RSA regime with $r \approx 10^{308}$ (RSA-1024), it fails in principle.

RSA as a Geometric Shield

In the FFGFT perspective, the difficulty of RSA factorisation is not a lack of computing power but a **shield of spacetime geometry**:

Large primes generate such long-period resonances in the feedback system $a^x \bmod N$ that no physical system — which is itself part of this spacetime and itself described by T_{rev} and ξ — can maintain the necessary phase stability over the runtime $r \cdot T_{\text{rev}}$.

The Wall Between Mathematics and Physics

Mathematics says: $N = p \cdot q$ is trivially true.

Physics says: To *discover* this truth, a system must sustain coherence for a period of $r \approx N$ cycles. For RSA-1024 this means $r \approx 10^{308}$ cycles at precision $\epsilon \approx 10^{-617}$ per cycle.

That is the wall. Not made of metal — made of time and phase.

.4 The Positive Result: Resonance Points Improve Error Correction

Preferred Resonance Points in the Torus Spectrum

From the same geometry that limits RSA decomposition follows a positive result for quantum error correction. The torus spectrum is not uniform: modes contribute as $1/n^2$, and prime modes are the irreducible resonance nodes. This means there are **preferred, geometrically stable positions** in the state space of the quantum computer.

Standard correction (threshold theorem) treats all errors statistically and projects onto an arbitrary target state. FFGFT-informed error correction recognises that certain states are **geometrically preferred** — they correspond to stable torus modes to which the system naturally returns.

Resonance Correction Instead of Statistical Correction

Rather than correcting every error independently, one can interpret errors as **drifting from stable resonance points**. Correction then consists of actively returning the system to the nearest stable resonance node — not to an arbitrary codeword state.

Improvement Potential Through Resonance Correction

When error correction is geometrically informed — using preferred torus modes as anchor points — it can correct errors more efficiently than the statistical method. The rate at which errors accumulate remains the same; but the rate at which they are corrected increases, because the corrective moves point toward natural attractors of the system.

This is not a contradiction of the fundamental runtime barrier: for small N , resonance correction helps considerably. For RSA key lengths, the period r exceeds the reach of any correction.

Practical Consequences for Willow and Successors

Google's Willow demonstrates exponential error suppression with growing code distance d (Doc. 183) — this is experimental evidence that topologically stable modes are accessible. The FFGFT perspective says:

- **What Willow already does:** Projection onto topologically stable torus modes — implicitly, without knowing the geometric structure.
- **What resonance correction could achieve:** Explicit use of the $1/n^2$ structure of the torus spectrum to direct corrections toward the most stable modes.
- **What remains in principle impossible:** Breaking RSA keys in the cryptographically relevant range — because the period $r \approx N$ exceeds the coherence time of any physical hardware.

.5 Counter-Arguments and FFGFT Responses

Standard Argument	FFGFT Response
Threshold theorem: QEC defeats time	QEC is itself a recursive process with runtime. For $r \approx 10^{308}$, correction becomes part of the noise.
Shor is polynomial: $O((\log N)^3)$ steps	Polynomiality holds for logical complexity. Phase precision $\epsilon \sim 1/N^2$ grows exponentially — physically, not logically.
Better hardware $\Rightarrow$ smaller ϵ	There is no $\epsilon = 0$. All hardware is embedded in spacetime, thus described by T_{rev} and ξ . Geometric phase drift is intrinsic.
Topological qubits solve the problem	Topological qubits improve error rates — they do not eliminate runtime accumulation. For RSA-1024, even $\epsilon \rightarrow 10^{-16}$ is insufficient; one would need 10^{-617} .

Table 2: Standard arguments of quantum information theory and FFGFT responses

.6 Summary

Time is the toll paid when mathematics is brought into physical reality. For small numbers this toll is affordable — quantum computers can and will break short-key cryptography. For RSA keys of present and future cryptography the toll is astronomical.

Three Statements of This Document

1. **Physical limit:** RSA-1024 and larger remains protected from quantum computers by the runtime accumulation of spacetime geometry (T_{rev}, ξ) — not by technical deficiencies, but by laws of nature.
2. **Static knowledge vs. dynamic process:** Known primes cost no runtime. Applying them to a new number N requires a physical phase coincidence — and that costs $r \cdot T_{\text{rev}}$.
3. **Positive result:** Error correction can be considerably improved through geometrically informed resonance correction (using preferred torus modes) — for small N , short periods, and practical quantum algorithms in the reachable range.

Draft. The statement on the physical impossibility of RSA decomposition at large key lengths is qualitatively motivated from FFGFT dynamics. A quantitative derivation of the precise scaling limit requires further formalisation of the connection between T_{rev} , ξ , and the threshold condition of Shor's algorithm.

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